MIL-T-6853(Aer)
NOTICE NO. 1
25 March 1960

MILITARY SPECIFICATION

TRANSMITTING SET, RADIO AN/ART-28

This notice forms a part of Specification MIL-T-6853(Aer) dated 1 August 1950 and shall constitute the covering page of the Specification.

Specification MIL-T-6853(Aer) dated 1 August 1950, and Amendment-1 dated 15 January 1956 are hereby downgraded from Confidential to UNCLASSIFIED

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MIL-T-6853(Aer)
1 August 1950

MILITARY SPECIFICATION

TRANSMITTING SET, RADIO, AN/ART-28

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be primarily designed to be an airborne radar relay transmitter to receive radar information from the AN/APS-20 Series Radar Sets or the AN/ARR-27 Radar Relay Receiver and to transmit this information to a remote AN/ARR-27 Radar Relay Receiver or to the AN/SRR-4; or AN/SRR-6 shipboard radar relay receiving equipments. This equipment shall generate the radio frequency carrier, accept the coded radar information, amplitude modulate the carrier with this information, and radiate the modulated signal. The modulation information shall be composed of video, IFF and azimuth sync information (azimuth sync information shall be defined as a triple pulses of basic, sine, cosine and main bang pulses of the type used in the SN-22A/APS-20 Synchronizer coding circuits) as supplied by a search radar or the AN/ARR-27 Radio Receiving Set.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Major Unit	Type Designation	Size (JAN-C-172)	Max. allow. Weight Lbs.	Applicable Paragraphs
Transmitter	T-271/ART-28	H1-D2	40	3.3.1
Power Supply	PP-615/ART-28	A1-D	25	3.3.2
Mountings				
Mounting Base	MT-628/ART-28	H1-D1 (Modified)		3.3.1.4
Mounting Base	MT-629/ART-28	A1-D		3.3.2.4

1.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

Contract - The contract shall have precedence over any specification.

This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Bureau of Aeronautics.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS AND DRAWINGS

2.1 GENERAL: The following specifications, publications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

2.2 SPECIFICATIONS:

Military Specifications -

JAN-C-172	Cases and Mounting Bases, Radio, Aircraft.
JAN-P-656	Packaging and Packing of Electrical Equipment and Spare Parts.

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MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment.
MIL-T-5422(Aer)	Testing; Environmental, Aircraft Electronic Equipment.

Air Force-Navy Aeronautical Specifications -

AN-E-19	Aircraft Electronic Equipment, General Specifications for.
AN-C-176	Control Panel; Aircraft Equipment, Rack or Console Mounted.

Bureau of Ships Specification -

RE-13A516	Nameplates

2.3 Other Publications -

Bureau of Aeronautics Publications -

NAVAER-2544	Power Requirements.
NAVAER-2586	Electronic Equipment Tube Complement.

2.4 Drawings -

Bureau of Aeronautics Drawing -

49A1B12	Input Power Connections, 28V DC and 115V AC Three Phase.,

2.5 AVAILABILITY OF SPECIFICATIONS, PUBLICATIONS AND DRAWINGS:

2.5.1 When requesting specifications, publications and drawings refer to both title and number.

2.5.2 Copies of Military and Air Force-Navy Aeronautical Specifications may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

2.5.3 Copies of Bureau of Ships Specifications may be obtained upon application to the Bureau of Ships, Electronics Division, Navy Department, Washington 25, D. C.

2.5.4 Copies of applicable Bureau of Aeronautics Publications and Drawings may be obtained upon application to Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attn: Electronics Division.

2.5.5 Copies of NAVAER-2586 and NAVAER-2544 may be obtained upon application to Supply Officer, Naval Air Supply Depot, 700 Robbins Avenue, Philadelphia, Pennsylvania, Attn: Publications Division.

3. REQUIREMENTS

3.1 Material and Workmanship -

3.1.1 General - The materials, component parts, mechanical assemblies and workmanship used in the construction of this equipment shall be in accordance with the Specification AN-E-19.

3.1.1.1 Electronic Equipment Tube Complement - All tube complements shall be reported by the contractor to the Bureau of Aeronautics by completing NAVAER-2586 in accordance with the instructions contained therein.

3.1.2 Detailed Parts List - A detailed parts list consisting of Group F and R parts including cable connectors, adaptors, coupling nuts, ferrules, wire, cable, and conduit with their respective manufacturing part numbers and weights, shall be prepared and submitted to the Bureau of Aeronautics upon approval of the production sample. Group F parts are parts fastened rigidly to the aircraft and group R parts are parts readily removable from the aircraft.

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3.2 General Design Requirements -

3.2.1 General - This equipment shall meet all the general design requirements as specified in Specification AN-E-19, unless specifically modified herein.

3.2.2 Total Weight - The total weight of the equipment shall not be greater than 80 pounds.

3.2.3 Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.4 Operating Life -

3.2.4.1 The equipment shall have a reliable operating life of at least 200 hours without requiring removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life should be specified by the manufacturer.

3.2.4.2 The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.5 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using unshielded interconnecting cabling. A small percentage of the individual wires may be shielded, if necessary, provided the assembly of the cable to its plugs may be easily accomplished. Reasonable precautions (such as the inclusion of a filter in the equipment) shall be taken to prevent the unshielded wires of the inter-connecting cables from producing fields of sufficient intensity to interfere with the satisfactory performance of the other electronic equipments normally installed in Naval Aircraft.

3.2.6 Cables and Connectors - The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification AN-E-19; however, cables and connectors shall not be supplied as part of the equipment.

3.2.7 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions specified in Specification AN-E-19.

3.2.8 Shock and Acceleration - The equipment shall meet the requirements of Specification AN-E-19 in regard to shock except that these requirements are modified to require that the equipment shall not be damaged or fail to provide the performance specified herein when subjected to impact shocks of 15G and shall be structurally capable of remaining in place when subjected to shocks of 30G.

3.2.9 Vibration Mounts - The vibration mounts shall meet the requirements of Specification JAN-C-172 unless modified herein. Mounts used which are modified versions of those specified in Specification JAN-C-172 or which are non-standard must meet the performance requirements of Specification JAN-C-172. Shock requirements of this specification apply in lieu of those in Specification JAN-C-172.

3.2.10 Power Requirements - The equipment shall operate on power from the following sources, with characteristics as defined in Specification AN-E-19 and with the exceptions as noted herein.

A. C. Source

115/200 Volts 320-1000 cps 3 phase 1000 VA

D. C. Source

28 Volts 3 AMPS

3.2.10.1 Voltage Limits - The equipment shall be designated to meet the performance requirements specified herein for the following voltage limits:

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3.2.10.1.1 D. C. Limits - The voltage limits of the D. C. Source shall be 24 to 29 volts.

3.2.10.1.2 A. C. Limits - The voltage limits of the A. C. Source shall be 110 to 120 volts line to neutral. In addition, the equipment shall operate at voltages within the limits of 102 to 110 and 120 to 124 but this operation may be at reduced performance and shall not damage the equipment. If practicable (i.e., no excess weight or form factor penalty) the equipment shall be designed to meet the performance requirements over the range of 102 to 124 volts.

3.2.10.2 A. C. Characteristics - The following A. C. characteristics shall apply in lieu of those shown in Specification AN-E-19.

(1) Voltage Limits - as specified in par. 3.2.10.1.2.
(2) Variation - not applicable.
(3) Transient - as shown on Fig. 1, except that the range of steady state line voltage and voltage limits shall be equal to the A. C. limits specified above.

3.2.10.3 Power Supply Frequency Deviation - The equipment shall be of such design that it will not be damaged by power supply frequencies below the minimum specified and shall automatically resume normal operation when the frequency returns within limits; however, there are no requirements on equipment performance at frequencies below the lower established operating limit. Down to 200 cycles the voltage shall be considered as maintained within the operating limits. Below 200 cycles the voltage shall be considered as varying proportional to the frequency, with zero voltage existing at zero frequency.

3.2.10.4 Power Input - The input power connections shall be in accordance with Bureau of Aeronautics Drawing 49A1B12 except that the following power "lines" shall be grounded internally with provisions for a single external ground:

a. The neutral of the three-phase power source.
b. The negative of the D. C. power source.

3.2.10.5 Power Source Transients - The equipment shall not be damaged when subjected to the applicable power source transients and shall automatically resume normal operation at the expiration of the transient condition.

3.2.10.6 Power Requirements Data - The contractor shall furnish power requirements data for the equipment on form NAVAER 2544. Preliminary and interim data shall be submitted concurrent with the development and design of the equipment. Final corrected data shall be submitted when the equipment design is completed. Changes in the design, which change the power requirements require the furnishing of corrected form NAVAER 2544.

3.2.10.7 Overload Protection - Overload protection for the equipment shall be provided in the equipment; all parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to any part of the equipment. The use of fuses and other protective devices shall be held to an absolute minimum.

3.2.10.8 Low Voltage Protection - Equipment shall not be damaged when voltages lower than the minimum specified operating voltages are applied.

3.2.11 Moisture and Fungus Resistant Treatment - Moisture and fungus resistant treatment shall be in accordance with the requirements of Specification AN-E-19 except the contractor may request a deviation from overall spraying of the equipment if parts and materials used are selected as non-nutrient for fungus or if the parts and materials are treated prior to their use in the equipment.

3.2.12 Nameplates - Nameplates shall be in accordance with Bureau of Ships Specification RE-13A516. The requirements for aircraft type nameplates shall apply.

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3.2.13 Interchangeability - The interchangeability requirements as specified in Specification AN-E-19 shall be extended to require interchangeability of units and component parts of this equipment with the latest model of any previously produced equipments. Such interchangeability shall be measured against a model, manufacturing drawings or other technical information as provided by the Bureau of Aeronautics for that purpose.

3.2.14 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of Specification AN-E-19.

3.2.15 Performance - The equipment shall be designed to meet the performance requirements as specified below:

3.2.15.1 Frequency - The equipment shall be tuneable within the frequency range of 460 to 510 mc/s.

3.2.15.2 Modulation Characteristics - The equipment shall be capable of transmitting information as contained in a coded pulse modulated carrier having a PRF of 300 per second. The envelope of the modulated carrier shall contain:

(1) Basic Pulse
(2) Sine Pulse
(3) Cosine Pulse
(4) Main Bang Pulse
(5) Radar, IFF, and Beacon Video Signals

The basic, sine, cosine, and main bang pulses are each composed of three one microsecond pulses whose leading edges are separated by 3 plus or minus 0.5 and 5 plus or minus 0.5 microseconds, respectively, as measured at 90 percent of peak amplitude.

3.2.16 Interference Limits - This equipment shall meet the requirements of Specification MIL-I-6181 with the exceptions and additions noted below.

3.2.16.1 Conducted - The requirements pertaining to conducted interference shall apply only to leads supplying power to the equipment.

3.2.16.2 Radiated - Measurements may be made with the equipment of the contractor's choice from the table of approved equipments given in Specification MIL-I-6181. The interference limits given in MIL-I-6181 for the particular combination of test equipment chosen will apply.

3.2.16.2.1 The radiated interference limits for the frequency range from 20 Mc. through 100 Mc. do not apply and measurements over this range may be omitted.

3.2.16.3 Test Arrangements - In addition to the general requirements of Specification MIL-I-6181, the equipment shall be assembled and operated so as to simulate an actual installation as closely as practicable; that is as to relative location of units, lengths of cables, use of dummy loads where required, etc. Shielded leads or cables should be used only where they will be so used in aircraft.

3.3 Detailed Design Requirements -

3.3.1 Radio Transmitter T-271/ART-28 - The transmitter shall be designed to meet the following requirements:

3.3.1.1 Function - The transmitter shall consist of an R.F. Assembly (including tuning assembly, oscillator and power amplifier), sync modulator, video modulator, monitor, tuning indicator, harmonic suppressing filter and video-sync signal test generator. It shall generate a radio frequency carrier within the frequency range of 460 - 510 Mc/s and modulate this carrier with sync and video inputs accepted from either the AN/ARR-27 Radar Relay Receiver or a search radar of the AN/APS-20 type to provide a modulated carrier to the antenna.

3.3.1.2 Size - The Radio Transmitter shall be of the size R1-D1 in accordance with Specification JAN-C-172.

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3.3.1.3 Weight - The weight of this unit, including mountings and plugs for cables, shall not exceed 45 pounds.

3.3.1.4 Mounting Provisions - This unit shall be equipped with a vibration mount of a type H1-D1 in accordance with Specification JAN-C-172. This mount shall be equipped with an attached cabling assembly compartment containing an assembly which consists of the necessary receptacles and/or connectors to receive sync, video, high voltage, 28 volts D.C. and all necessary power and control voltages, and to provide input connections to the transmitter.

3.3.1.5 Frequency - This unit shall be tuneable over the frequency range of 460 to 510 Mc/s and shall have provisions whereby any one of four channels may be quickly and easily selected for operation. Equipment delivered by the contractor shall be so adjusted as to have channel center frequencies of 464.87 Mc/s., 477.43 Mc/s., 490 Mc/s. and 502.56 Mc/s. for channels 1, 2, 3 and 4 respectively.

3.3.1.6 Input - This unit shall provide modulation of the r.f. carrier from two input signals of either positive or negative polarity; one a composite video-sync signal of 1.0 volt to a 200 ohm termination and the other a synchronizing signal of 5 volts to a 470 ohm termination.

3.3.1.7 Linearity - The ratio between any two signal amplitudes shall not vary more than plus or minus 10% as measured between the input to the transmitter and the output signal.

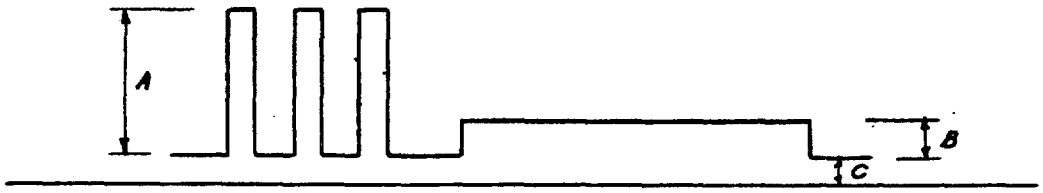
3.3.1.8 Power Output - The output power of the transmitter shall be as follows:

3.3.1.8.1 Peak Video Power - The output power as measured at the antenna receptacle shall be 200 watts or more on video peaks.

3.3.1.8.2 Peak Sync Power - The output power as measured at the antenna receptacle shall be 800 watts or more on sync peaks.

3.3.1.8.3 Sync-Video Ratio - The sync-video power ratio shall be four to one or greater and shall be defined as follows:

$$\text{Sync-Video Ratio} = \frac{A}{B}$$



3.3.1.8.4 Unmodulated Power - The unmodulated power "C" shall be adjusted to 20 watts.

3.3.1.9 Bandwidth - The transmitter shall have sufficient fidelity to pass a 2400 microsecond rectangular signal pulse with ninety-five percent of peak amplitude at the trailing edge, and all other modulating frequencies up to four megacycles without attenuation of more than 3 db.

3.3.1.10 Cooling System - The transmitter shall be air cooled by means of a suitable blower.

3.3.1.11 R-F Assembly - The R-F Assembly shall consist of an oscillator and a power amplifier with the necessary provisions for interstage coupling, tuning and adjustment controls, shielding and bypassing to prevent r-f leakage, and air cooling.

3.3.1.11.1 Oscillator - The oscillator shall be of the grounded grid type designed around the type 5588 vacuum tube and employing a cavity resonator. The cavity resonator shall be so designed as to be effectively a quarter-wave shorted line in the plate circuit and a half-wave open line in the cathode circuit.

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3.3.1.11.2 Power Amplifier - The power amplifier shall be designed around two type 4X150A vacuum tubes connected in parallel and operating in a single cavity resonator. The cavity resonator shall be so designed to provide effectively a condenser tuned quarter-wave shorted line in the plate circuit. The common grid shall be coupled to the oscillator by means of a coupling loop in the oscillator plate circuit and a series condenser tuned transmission line. Provisions shall be made for inductively coupling output power from the power amplifier to the antenna. There shall be incorporated in the output circuit a low-pass filter having a standing wave voltage ratio of not more than 1.5 in the 460-510 Mc/s band and having a negligible attenuation (less than 1 db) in this band and an attenuation of not less than 55 db for all frequencies above 920 Mc/s.

3.3.1.11.3 Tuning - Tuning of the transmitter through its frequency range shall be accomplished by means of control brought out to the front panel and so linked to provide means of tuning the oscillator cathode, oscillator plate, amplifier grids and amplifier plate circuits.

3.3.1.11.4 Monitor - A monitor circuit shall be provided that rectifies and presents in the form of a meter or scope presentation some of the r-f power output.

3.3.1.12 Tuning Indicator - Circuitry incorporating a crystal oscillator operating at frequencies of channel separation shall be provided and so designed that its operation causes an electron ray indicator tube to close to a minimum shadow when the transmitter is tuned to the channel frequencies.

3.3.1.13 Modulator - The modulator shall provide the necessary modulation voltages to modulate the R-F Power Amplifier with sync-video modulation on the grids and sync pulses on the plates. The modulator shall have a 2 Mc/s video bandwidth. A front panel polarity switch for accommodating either the AN/ARR-27 Radio Receiving Set or the AN/APS-20 Radar Set as a source of modulation signals shall be provided.

3.3.1.14 Video - Sync Signal Test Generator - Circuitry for a self contained signal generator shall be provided, the purpose of which is to provide a one-half duty cycle gate and simulated sync pulses when these are not provided by the radar synchronizer. A switch shall be provided which can be actuated by contact with the equipment test stand to allow automatic operation of this generator when the transmitter is being bench checked.

3.3.1.15 Relay Box - The relay box shall be an integral part of the transmitter and shall contain a series of relays as required to provide overload protection and visual front panel warning indication of overload in the oscillator plate circuit and for each amplifier plate circuit, overload recovery, and time delay circuits for the control of high voltages.

3.3.1.16 Front Panel - The front panel shall contain a tuning calibration chart, all visual indicators, power ON-OFF switch, AN/ARR-27 - AN/APS-20 input polarity switch, signal generator test switch, antenna and monitor receptacles and tuning controls.

3.3.1.16.1 Front Panel Meter Indicators - Two meters shall be provided on the front panel, one to give an indication of grid current magnitude and one to indicate power output to the antenna. The power output meter shall be so scaled to indicate sub-normal performance, required CW power, and normal power output during modulation by signals from the incorporated signal generator.

3.3.1.16.2 Front Panel Receptacles - The following receptacles shall be provided on the front panel:

3.3.1.16.2.1 Antenna Receptacle - This receptacle shall receive plug UG-204A/U and/or plug UG-27A/U and furnish r-f power to the antenna.

3.3.1.16.2.2 Monitor Receptacle - This receptacle shall receive plug UG-260/U and shall furnish a demodulated signal for purposes of viewing the signal as transmitted with an external scope.

3.3.1.16.3 Tuning Controls - Tuning controls shall be provided on the front panel for making tuning adjustments in the oscillator cathode, oscillator plate, amplifier grid, and amplifier plate circuits.

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3.3.1.17 Receptacles - The necessary plug and/or receptacle arrangement shall be provided at the rear of the transmitter to receive sync, video, high voltage and all necessary power and control voltages from the cabling assembly compartment described in paragraph 3.3.1.4.

3.3.2 Power Supply PP-615/ART-28 - The power supply shall be designed to meet the following requirements:

3.3.2.1 Function - This unit shall provide all power requirements of the transmitter unit.

3.3.2.2 Size - The power supply unit shall be of size A1-D in accordance with Specification JAN-C-172.

3.3.2.3 Weight - The weight of this unit shall not exceed 25 pounds.

3.3.2.4 Mounting Provisions - This unit shall be equipped with a vibration mounting MT-829/ART-28 in accordance with Specification JAN-C-172.

3.3.2.5 Voltages - This unit shall provide the following voltages necessary for proper operation of the transmitter unit:

(1) Positive 1000 Volts DC at 700 ma.
(2) Positive 300 Volts DC at 250 ma.
(3) Negative 200 Volts DC at 5 ma.
(4) Negative 120 Volts DC at 100 ma.

3.3.2.6 Power Supply Control - The operation of the power supply unit shall be automatically controlled by the ON-OFF switch on the front panel of the transmitter unit. This control is effected by a relay in the power unit which is actuated by 28 volts DC from the transmitter unit.

3.3.2.7 Receptacles - The following receptacles shall be provided on the front of the power supply unit:

3.3.2.7.1 High Voltage Receptacle - This receptacle shall furnish a positive 1000 volts DC for transmission to the transmitter unit.

3.3.2.7.2 Power Receptacle - This receptacle shall furnish the following voltages for use in the transmitter unit:

(1) Positive 300 Volts DC
(2) Negative 200 Volts DC
(3) Negative 120 Volts DC
(4) Positive 28 Volts DC - (Power Supply Control Voltage)
(5) Single Phase 115 Volt AC

3.3.2.7.3 Power Input Receptacle - This receptacle shall be capable of accepting the nominal 115-volt 320-1000 cps three phase AC Power, 28 volt DC Power from the aircraft main power source.

4. METHODS OF SAMPLING, INSPECTION AND TESTS -

4.1 Pre-Production Tests - Pre-production tests shall be conducted on a model representative of the production equipment to determine that the production equipment meets the requirements of this specification. Pre-production tests shall be conducted at a laboratory designated by the Bureau of Aeronautics, or when so stated in the contract at the contractor's plant under the supervision of the Bureau of Aeronautics.

4.1.1 Scope of Tests - Pre-production tests shall include all tests deemed necessary to determine that the model meets all the requirements of this specification and the contract. Pre-production tests shall include testing in accordance with Specification MIL-T-5422(Aer). Test data to be supplied by the contractor shall include data obtained by the contractor in testing to Specification MIL-T-5422(Aer).

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4.1.2 Accessory Material and Design Data - Accessory material and design data to be supplied with the pre-production model shall be in accordance with the requirements of Specification AN-E-19.

4.1.3 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief with all applicable requirements. The design data submitted by the contractor for the pre-production model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

4.1.4 Model Acceptance - Acceptance of the pre-production model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the pre-production model. Pre-fabrication of any equipments prior to the approval of the pre-production model is at the contractor's own risk. The approved pre-production model will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The pre-production model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2 Inspection Tests - Inspection tests shall be conducted on equipment submitted for acceptance under the contract to determine that it is equivalent in performance and construction to the approved pre-production model. Inspection tests shall be conducted at the contractor's plant under the supervision of the naval inspector, unless specified otherwise in the contract. Inspection tests shall meet all the requirements specified in Specification AN-E-19 under inspection tests.

4.2.1 Scope of Tests - Inspection tests shall consist of the following two tests:

4.2.1.1 Individual Tests - Individual tests shall be conducted on each equipment submitted for acceptance under the contract.

4.2.1.2 Sample Test - Sample Test shall be conducted on a quantity as approved by the Bureau of Aeronautics. Samples submitted for sample tests shall first have passed the individual tests. Sample tests shall be more extensive than the individual tests and may include any of the tests listed under pre-production tests or any other tests deemed necessary by the inspector to determine that the production equipment continues to be equivalent in performance and construction of the approved pre-production model.

4.3 Test Procedures - The procedures and methods for conducting both pre-production tests and inspection tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the naval inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed, and marked for shipment in accordance with the requirements of Specification JAN-P-658.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 26.5 volts D C and 115 volts A C. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

Patent Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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